IT Ticket System

Enterprise Service Management Platform - Documentation & Implementation Guide

Table of Contents

1. System Overview
2. System Architecture
3. Implementation Workflow
4. Service Structure Configuration
5. Support Teams Management
6. Business Rules Engine
7. Workflow Automation
8. SLA Configuration
9. System Output & Integration
10. Business Benefits

1. System Overview

The IT Ticket System is an enterprise-grade service management platform that automates ticket classification, routing, and resolution workflows. Built on a powerful business rules engine, the system eliminates manual processes and ensures consistent service delivery across your organization.

Core Capabilities

- Automated ticket classification and categorization
- Intelligent routing to appropriate teams and agents
- Dynamic SLA policy application and monitoring
- Multi-level approval workflows
- Automatic escalation management
- Comprehensive audit trail and reporting

Key Value Proposition

Upon ticket submission, the system automatically analyzes the request and returns a comprehensive JSON response containing all necessary routing, SLA, and workflow information. This enables seamless integration with your existing development infrastructure.

2. System Architecture

```
User Submission
↓
┌───────────────────────────────────┐
│ Classification Engine │
│ • Service Group Detection │
│ • Ticket Type Identification │
│ • Issue Type Mapping │
└───────────────────────────────────┘
↓
┌───────────────────────────────────┐
│ Business Rules Engine │
│ • Routing Rules │
│ • Approval Rules │
│ • Custom Field Validation │
└───────────────────────────────────┘
↓
┌───────────────────────────────────┐
│ SLA & Workflow Engine │
│ • SLA Policy Application │
│ • Workflow State Management │
│ • Escalation Rules │
└───────────────────────────────────┘
↓
JSON Response → API Integration
```

3. Implementation Workflow

The system configuration follows a structured, sequential approach to ensure all dependencies are properly established.

1. **Service Structure Configuration** - Define service groups, ticket types, and issue types to establish your classification taxonomy
2. **Team Structure Setup** - Create support teams with appropriate levels and assign members
3. **Business Rules Definition** - Configure routing, approval, escalation, and custom field rules
4. **Workflow Configuration** - Define ticket lifecycle states and transitions
5. **SLA Policy Setup** - Establish service level agreements and response/resolution targets
6. **Testing & Validation** - Verify configuration through test scenarios

4. Service Structure Configuration

4.1. Service Groups

Service Groups represent high-level service categories that organize your support operations by department or functional area.

Service Group	Code	Description	Scope
IT Infrastructure	IT_INFRA	Infrastructure and systems	Servers, networks, hardware
Software & Applications	SOFTWARE	Application support	Business applications, licenses
Human Resources	HR	HR services	Leave, payroll, onboarding
Facilities Management	FACILITIES	Office and facilities	Office equipment, maintenance

4.2. Ticket Types

Ticket Types define the nature of requests within each Service Group, enabling more granular classification.

Ticket Type	Code	Service Group	Typical Use Cases
Access Request	ACCESS_REQ	IT Infrastructure	VPN, system access, permissions
Hardware Issue	HW_ISSUE	IT Infrastructure	Equipment failures, replacements
Network Problem	NET_ISSUE	IT Infrastructure	Connectivity, performance issues
Software Installation	SW_INSTALL	Software & Applications	Application installation, updates

4.3. Issue Types

Issue Types provide the most granular level of classification, supporting hierarchical structures with parent-child relationships.

Example: Access Request Hierarchy

```
Access Request
├── VPN Access (VPN_ACCESS)
│ ├── New VPN Account (VPN_NEW)
│ ├── VPN Connection Issue (VPN_ISSUE)
│ └── VPN Account Termination (VPN_TERM)
├── Email Access (EMAIL_ACCESS)
│ ├── New Email Account (EMAIL_NEW)
│ ├── Password Reset (EMAIL_RESET)
│ └── Mailbox Quota Increase (EMAIL_QUOTA)
└── System Access (SYS_ACCESS)
├── ERP Access Request (ERP_ACCESS)
├── CRM Access Request (CRM_ACCESS)
└── Database Access (DB_ACCESS)
```

5. Support Teams Management

Support teams are organized in a tiered structure (L1, L2, L3) to facilitate efficient escalation and resolution processes.

Team Name	Level	Service Group	Responsibilities
L1 IT Support	Level 1	IT Infrastructure	First-line support, basic troubleshooting
L2 Network Team	Level 2	IT Infrastructure	Advanced network diagnostics and resolution
L3 Infrastructure	Level 3	IT Infrastructure	Expert-level infrastructure support

6. Business Rules Engine

6.1. Routing Rules

Routing Rules automatically assign tickets to appropriate teams based on issue classification and business logic.

Rule Name	Issue Type	Assigned To	Priority
VPN Access Routing	VPN Access	L1 IT Support	High
Network Issue Routing	Network Problem	L2 Network Team	Critical
Hardware Replacement	Hardware Issue	L1 IT Support	High

6.2. Approval Rules

Approval Rules define required approvals before ticket processing, supporting multi-level approval chains.

Approval Workflow Example: Budget Request

```
Request Amount > $10,000:

Step 1: Direct Manager
↓ (Approved)
Step 2: Department Head
↓ (Approved)
Step 3: Finance Director
↓ (Approved)
Step 4: CEO (Final Approval)
↓ (Approved)
Process Request → Complete
```

Rule Name	Issue Type	Approver Level	Condition
Leave Approval	Leave Request	Direct Manager	Always required
System Access Approval	System Access	IT Manager	If accessing >3 systems
Budget Approval	Budget Request	Finance Director	If amount >$5,000

6.3. Escalation Rules

Escalation Rules ensure timely resolution by automatically escalating tickets that exceed defined time thresholds.

Level	Team	Time Limit	Escalation Action
L1	L1 IT Support	4 hours	Basic troubleshooting and diagnosis
L2	L2 Network Team	8 hours	Advanced technical investigation
L3	L3 Infrastructure	24 hours	Expert-level resolution or vendor escalation

Automatic Escalation Process

When time limits are exceeded:

- Ticket automatically reassigned to next level team
- Email notifications sent to both current and new team
- Priority may be elevated based on configuration
- Escalation logged in ticket audit trail

6.4. Custom Fields Configuration

Custom Fields enable collection of issue-specific information, ensuring agents have all necessary data upfront.

Field Name	Type	Required	Validation Rules
VPN Username	Text Input	Yes	Alphanumeric only, 3-20 characters
Department	Dropdown	Yes	Predefined list
Business Justification	Textarea	Yes	Minimum 20 characters
Access Duration	Number	No	1-365 days

9. System Output & Integration

9.1. User Submission Example

```
User Input:
  Service Group: IT Infrastructure
  Ticket Type: Access Request
  Issue Type: VPN Access → New VPN Account
  VPN Username: john.doe
  Department: Engineering
  Justification: Required for remote work access to development servers
```

9.2. System JSON Response

```
{
  "ticket_id": "TKT-2024-001234",
  "status": "submitted",
  "created_at": "2024-12-14 10:30:00",

  "user": {
    "id": 12345,
    "name": "John Doe",
    "email": "john.doe@company.com",
    "department": "Engineering"
  },

  "classification": {
    "service_group": {
      "id": 1,
      "code": "IT_INFRA",
      "name": "IT Infrastructure"
    },
    "ticket_type": {
      "id": 5,
      "code": "ACCESS_REQ",
      "name": "Access Request"
    },
    "issue_type": {
```

9.3. Developer Integration Examples

Automatic Assignment Implementation

```
// PHP Laravel Example
$response = TicketSystem::submit($ticketData);
$ticketId = $response['ticket_id'];

// Auto-assign to team
Ticket::create([
    'ticket_id' => $ticketId,
    'assigned_team_id' => $response['routing']['assigned_to']['id'],
    'priority' => $response['routing']['priority'],
    'status' => $response['status']
]);

// Set SLA deadlines
SLA::create([
    'ticket_id' => $ticketId,
    'first_response_deadline' => $response['sla']['first_response']['deadline'],
    'resolution_deadline' => $response['sla']['resolution']['deadline']
]);
```

Approval Workflow Implementation

```
// Handle approval routing
if ($response['approval']['required']) {
    foreach ($response['approval']['approvers'] as $approver) {
        Approval::create([
            'ticket_id' => $ticketId,
            'approver_id' => $approver['user_id'],
            'level' => $approver['level'],
            'status' => 'pending'
        ]);

        // Send notification
        Notification::send($approver['user_id'], [
            'type' => 'approval_request',
            'ticket_id' => $ticketId
        ]);
    }
}
```

SLA Monitoring Implementation

```
// Background job for SLA monitoring
public function checkSLABreach() {
    $tickets = Ticket::where('status', '!=', 'closed')
                     ->with('sla')
                     ->get();

    foreach ($tickets as $ticket) {
        if (now() > $ticket->sla->first_response_deadline
            && !$ticket->first_response_sent) {
            // Breach detected - notify management
            $this->notifySLABreach($ticket, 'first_response');
        }

        if (now() > $ticket->sla->resolution_deadline
            && $ticket->status != 'resolved') {
            // Resolution SLA breach
            $this->notifySLABreach($ticket, 'resolution');
        }
    }
}
```

Auto-Escalation Implementation

```
// Automated escalation based on time limits
public function checkEscalation($ticketId) {
    $ticket = Ticket::with(['escalation', 'routing'])->find($ticketId);
    $currentLevel = $ticket->routing->level;
    $timeInLevel = now()->diffInHours($ticket->last_level_change);

    $escalationRule = $ticket->escalation->levels
        ->where('level', $currentLevel)
        ->first();

    if ($timeInLevel >= $escalationRule->time_limit_hours) {
        $nextLevel = $currentLevel + 1;
        $nextTeam = $ticket->escalation->levels
            ->where('level', $nextLevel)
            ->first();

        // Escalate ticket
        $ticket->update([
            'assigned_team_id' => $nextTeam->team_id,
            'level' => $nextLevel,
            'last_level_change' => now()
        ]);

        // Notify both teams
        $this->notifyEscalation($ticket, $currentLevel, $nextLevel);
    }
}
```

7. Workflow Automation

Workflows define the ticket lifecycle, including states, transitions, and business rules for state changes.

7.1. Standard Workflow States

State	Description	Allowed Transitions
New	Ticket just created	In Progress, Pending Approval
Pending Approval	Awaiting approval	Approved, Rejected
In Progress	Being worked on	Resolved, On Hold, Escalated
On Hold	Waiting for information	In Progress, Cancelled
Resolved	Solution provided	Closed, Reopened
Closed	Ticket completed	Reopened (conditional)

7.2. Workflow Transition Rules

Example: VPN Access Request Workflow

```
New
↓
Pending Approval (IT Manager)
↓ (Approved)
In Progress (L1 IT Support)
↓
Resolved (VPN account created)
↓
Closed (User confirmed access)
```

7.3. Conditional Transitions

From State	To State	Condition	Action
In Progress	Escalated	Time limit exceeded	Auto-assign to L2 team
Resolved	Closed	User confirmation received	Update SLA metrics
On Hold	Cancelled	No response for 7 days	Send final notification

8. SLA Configuration

Service Level Agreements define response and resolution time commitments based on ticket priority and type.

8.1. SLA Policies

Policy Name	Priority	First Response	Resolution Time
Critical SLA	Critical	30 minutes	4 hours
High Priority SLA	High	2 hours	8 hours
Medium Priority SLA	Medium	4 hours	24 hours
Low Priority SLA	Low	8 hours	72 hours

8.2. SLA Assignment Rules

SLA policies are automatically assigned based on issue type classification and business rules.

```
SLA Assignment Logic:

IF issue_type = "Network Problem" → Critical SLA
IF issue_type = "VPN Access" → High Priority SLA
IF issue_type = "Software Installation" → Medium Priority SLA
IF issue_type = "General Inquiry" → Low Priority SLA

Custom Rules:
- VIP users → Upgrade SLA by one level
- After-hours tickets → Extended SLA (×1.5)
- Holidays → Extended SLA (×2.0)
```

8.3. SLA Monitoring & Alerts

Alert Type	Trigger Point	Recipients	Action
Warning Alert	75% of SLA time elapsed	Assigned agent, Team lead	Priority notification
Critical Alert	90% of SLA time elapsed	Agent, Manager, Director	Escalation preparation
Breach Alert	SLA deadline exceeded	All stakeholders	Incident report generated

SLA Breach Management

When SLA is breached:

- Automatic notification to management chain
- Incident report generated with root cause analysis
- Performance metrics updated for team/agent
- Optional automatic escalation to next level

10. Business Benefits

10.1. Operational Efficiency

Metric	Before Implementation	After Implementation	Improvement
Average Assignment Time	15-30 minutes	Instant (automated)	95% faster
SLA Compliance Rate	60-70%	90-95%	+30% improvement
First Response Time	4-6 hours	1-2 hours	70% faster
Resolution Time	48-72 hours	12-24 hours	65% faster
Ticket Misrouting Rate	20-25%	2-5%	85% reduction

10.2. Key Business Value

- **Reduced Manual Effort:** Automated classification and routing eliminates 80% of manual triage work
- **Improved User Satisfaction:** Faster response times and consistent service delivery
- **Better Resource Utilization:** Intelligent routing ensures optimal workload distribution
- **Enhanced Compliance:** Comprehensive audit trails and approval workflows
- **Data-Driven Insights:** Rich analytics from structured ticket data
- **Scalability:** Handle increasing ticket volumes without proportional staff increase

10.3. ROI Analysis

Typical Return on Investment

Cost Savings (Annual):

- Reduced administrative overhead: 40-50% time savings
- Decreased escalation costs: 30% reduction in senior staff involvement
- Lower SLA penalty costs: 85% reduction in breach incidents
- Improved productivity: 25-35% increase in tickets resolved per agent

Implementation Timeline: 4-8 weeks for full deployment

Payback Period: Typically 6-9 months

10.4. Comparative Analysis

Aspect	Manual Process	Automated System
Ticket Assignment	Manual review and assignment	Instant automated routing
Priority Setting	Subjective, inconsistent	Rule-based, consistent
SLA Tracking	Manual tracking, prone to oversight	Real-time monitoring with alerts
Escalation	Manual identification and routing	Automatic escalation based on rules
Approval Process	Email chains, delayed responses	Automated workflow with notifications
Reporting	Manual data compilation	Real-time dashboards and analytics

11. Implementation Checklist

Phase 1: Foundation Setup

1. Define Service Groups (2-4 primary groups)
2. Create Ticket Types for each Service Group (3-5 types per group)
3. Define Issue Types with hierarchical structure (5-10 issues per ticket type)
4. Document classification taxonomy

Phase 2: Team Structure

1. Create Support Teams with level assignments (L1, L2, L3)
2. Assign team members and define roles
3. Map teams to Service Groups
4. Define escalation paths between teams

Phase 3: Business Rules

1. Configure Routing Rules (minimum 1 rule per issue type)
2. Setup Approval Rules for sensitive requests
3. Define Escalation Rules with time limits
4. Create Custom Fields for data collection
5. Test rule logic with sample scenarios

Phase 4: Workflow & SLA

1. Define workflow states and transitions
2. Configure SLA policies based on priority
3. Setup SLA assignment rules
4. Configure monitoring and alert thresholds

Phase 5: Testing & Validation

1. Create test scenarios covering all issue types
2. Validate routing accuracy
3. Test approval workflows
4. Verify SLA calculations
5. Test escalation mechanisms
6. Conduct user acceptance testing

Phase 6: Deployment

1. Train support teams on new system
2. Conduct end-user training sessions
3. Deploy to production environment
4. Monitor initial performance
5. Collect feedback and iterate

12. Best Practices

12.1. Classification Design

- Keep Service Groups broad and stable (rarely change)
- Use descriptive, user-friendly names for Ticket Types
- Limit Issue Type hierarchy to 2-3 levels maximum
- Review and refine classification quarterly based on usage patterns

12.2. Routing Rules

- Start with basic rules, add complexity as needed
- Avoid overlapping rules that may cause conflicts
- Document rule logic for future reference
- Review routing accuracy monthly and adjust rules

12.3. SLA Management

- Set realistic SLA targets based on historical data
- Build in buffer time for complex issues
- Consider business hours vs. calendar hours
- Account for holidays and special circumstances

12.4. Change Management

- Communicate system changes to all stakeholders
- Provide adequate training before major updates
- Maintain version control for rule configurations
- Document all configuration changes with rationale

13. System Integration Points

13.1. API Endpoints

Endpoint	Method	Purpose
/api/tickets/submit	POST	Submit new ticket and receive routing/SLA data
/api/tickets/{id}/status	PUT	Update ticket status
/api/tickets/{id}/escalate	POST	Manual escalation trigger
/api/approvals/{id}/action	POST	Approve or reject ticket

13.2. Webhook Events

- **ticket.created** - New ticket submitted
- **ticket.assigned** - Ticket assigned to team/agent
- **ticket.escalated** - Ticket escalated to higher level
- **sla.warning** - SLA warning threshold reached
- **sla.breached** - SLA deadline exceeded
- **approval.required** - Approval request sent
- **ticket.resolved** - Ticket marked as resolved
- **ticket.closed** - Ticket closed

© 2024 - Confidential & Proprietary